

- Tabelas-Verdade e expressões Booleanas

1.

A	B	C	F
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

C \ AB	00	01	11	10
0	0	0	1	0
1	1	1	1	1

$$F = C + AB$$

3.

A	B	C	D	H
0	0	0	0	1
0	0	0	1	1
0	0	1	0	1
0	0	1	1	1
0	1	0	0	0
0	1	0	1	0
0	1	1	0	0
0	1	1	1	0
1	0	0	0	1
1	0	0	1	1
1	0	1	0	1
1	0	1	1	1
1	1	0	0	0
1	1	0	1	0
1	1	1	0	1
1	1	1	1	1

C \ AB	00	01	11	10
00	1	0	0	1
01	1	0	0	1
11	1	0	1	1
10	1	0	1	1

$$H = \bar{B} + AC$$

- Simplificações com álgebra booleana

6. $F = AB + \cancel{A\bar{B}} + \cancel{\bar{A}B}$
 $F = AB$

9. $F = \cancel{\bar{A}\bar{B}} + \cancel{A\bar{B}} + \bar{A}\bar{B}C$
 $F = \bar{A} + \bar{B}C$

12. $F = AB + \bar{A}C + \bar{B}C$
 $F = AB + \bar{A}C$

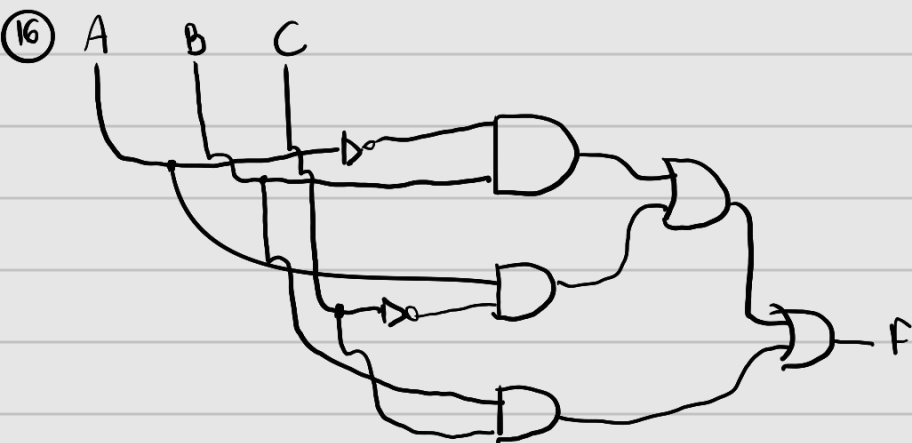
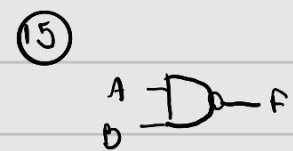
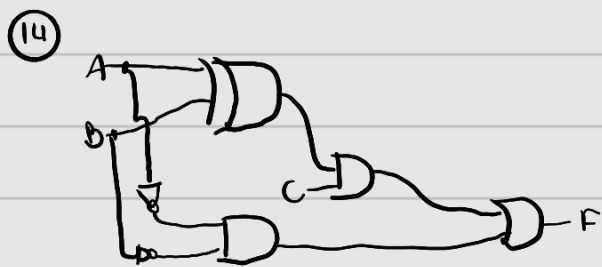
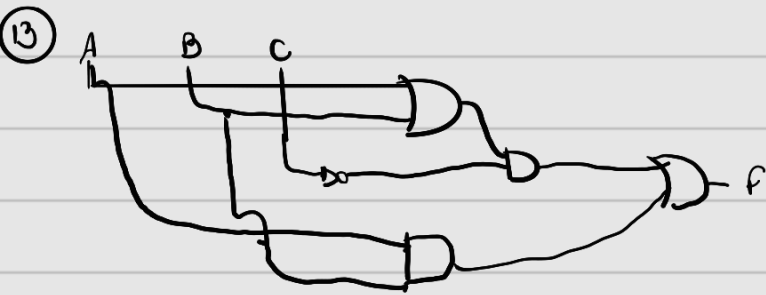
7. $F = (A + \bar{B})(A + \bar{B})$
 $F = A$

10. $F = \overbrace{A + \bar{A}\bar{B}}^{A + \bar{A}}(\bar{A} + B)$
 $F = A + B$

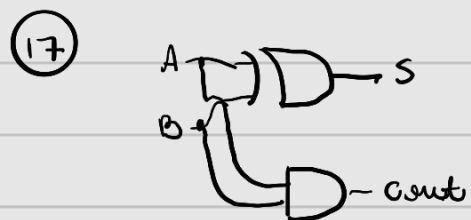
8. $F = \cancel{A\bar{B}C} + \cancel{A\bar{B}C} + \cancel{AC\bar{B}} + \cancel{AC\bar{B}}$
 $F = AC$

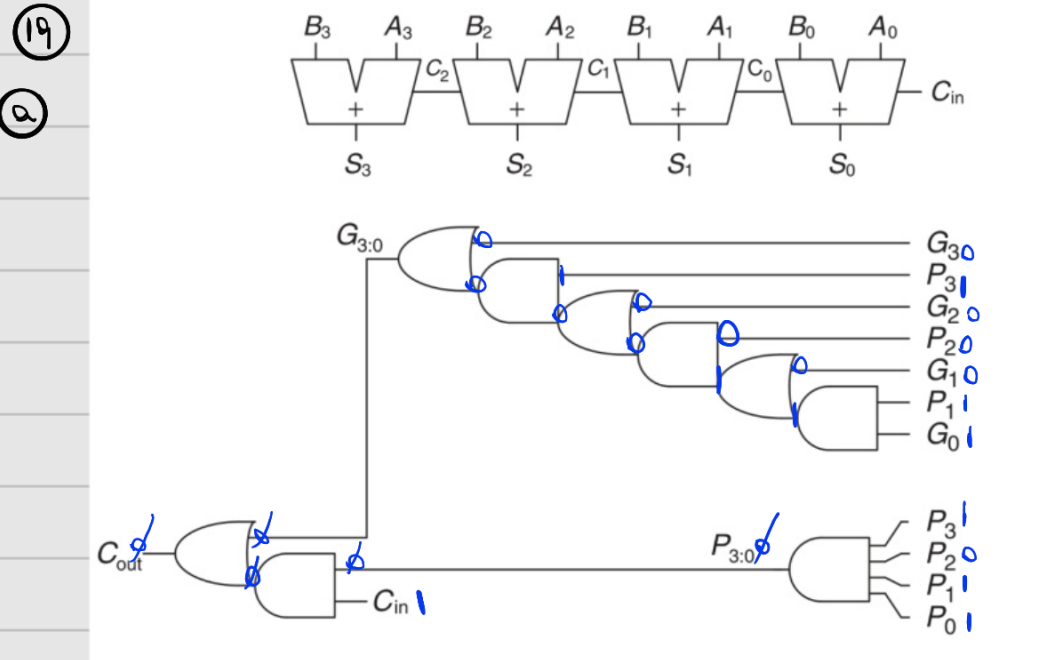
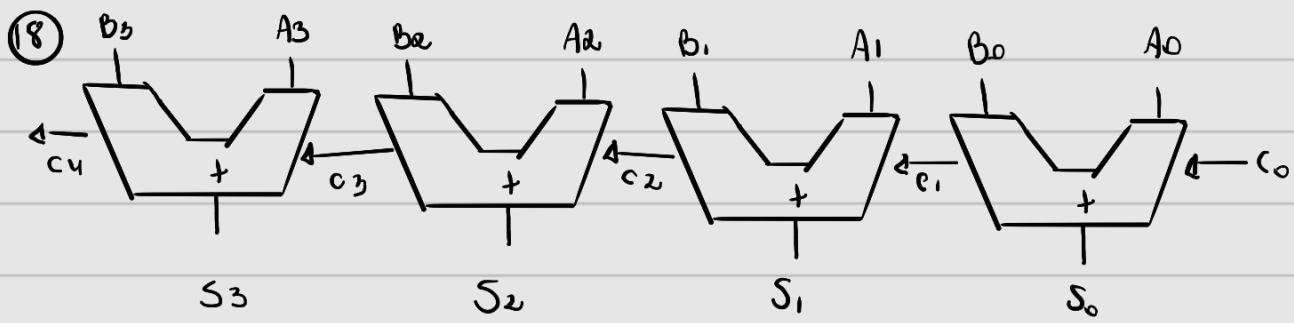
11. $F = (\cancel{A + \bar{B} + C})(\cancel{A + \bar{B} + C})$
 $F = A + C$

- Expressões para circuitos

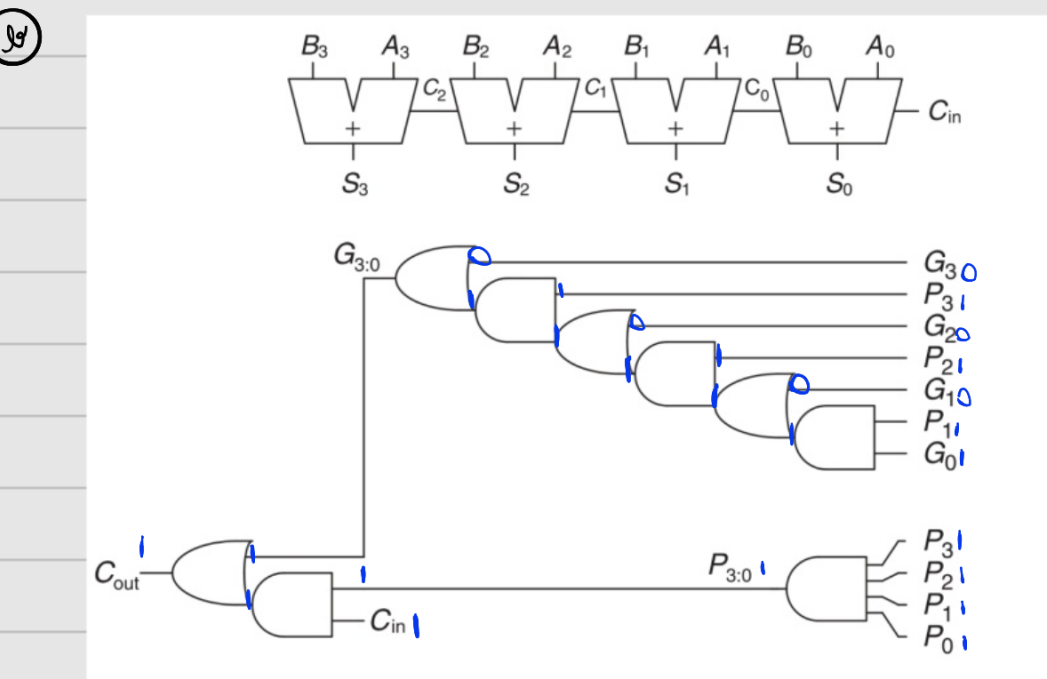


- Semáforos





$$\begin{array}{r} 3210 \\ 1011 \\ + 0001 \\ \hline \end{array} \quad C_{in}=1$$



$$\begin{array}{r} 3210 \\ 1011 \\ + 0101 \\ \hline \end{array} \quad C_{in}=1$$

- Circuitos para expressões

Q0 $Y = A\bar{B} + AB + AC$
 $Y = A$

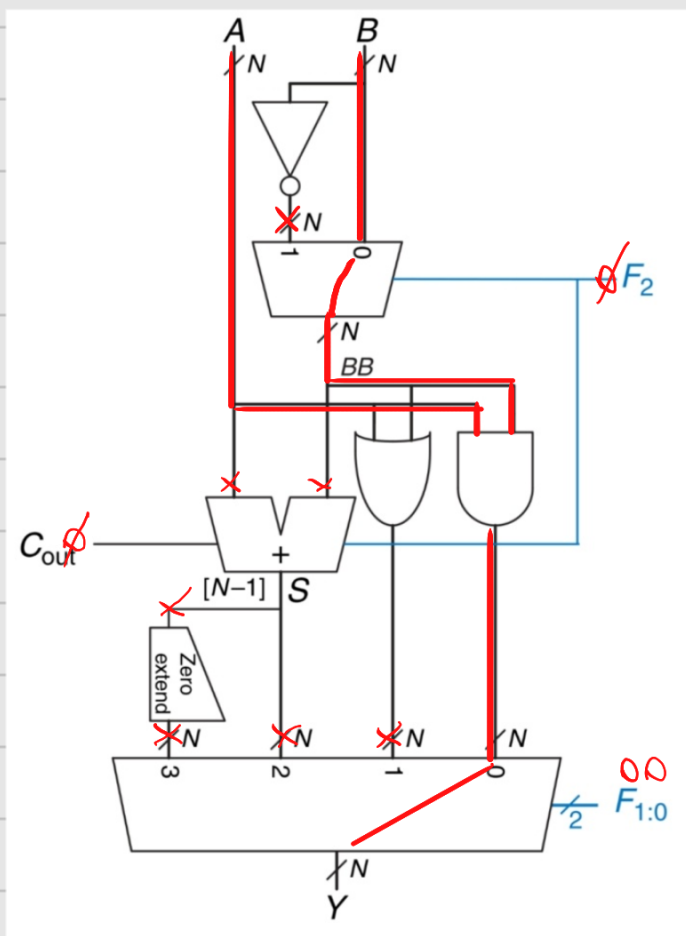
Q1 $G = A\bar{C} + BC + AB$
 $G = A + BC$

Q2 $X = A \oplus 0$

- Cilu

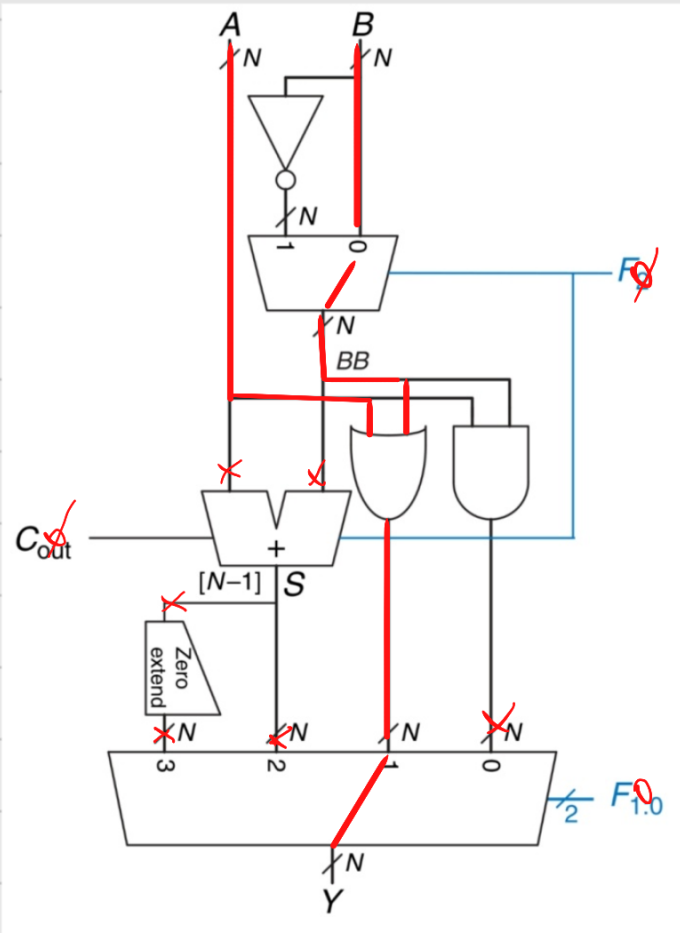
Q3

Instrução AND



$Y = AB$

Instructions or



$$y = A + B$$